

Gathering Requirements

- 1. Identify the technical and functional requirements for the 3D models.**
- 2. Analyze the project needs regarding detail quality, materials, lighting, and animation.**

1. Identify the Technical and Functional Requirements for the 3D Models

- **File Format:** 3d Max
- **Modeling Software:** 3ds Max that will be used to create the 3D models.
- **Models:** perfume bottle , packages, curtain , flower petals and texts

2. Analyze the Project Needs Regarding Detail Quality, Materials, Lighting, and Animation

- **Detail Quality ,Design Complexity and Material:**

- 1-bottle consist of two parts above part is glass (clear colored glass and transparent) and lower part is surrounded by plastic part.
- 2-Fine mist nozzle is made of metal.
- 3-Cap part is made of plastic part.
- 4-The liquid (colored liquid and transparent).

2. Analyze the Project Needs Regarding Detail Quality, Materials, Lighting, and Animation

- **Lighting:**(Arnold light)
 - **Lighting Setup:** Plan the lighting setup for rendering. This may include the Area Light or Spotlight from the Arnold lights. to create a visually appealing presentation of the bottle.
 - Key light :This light should be the primary light source illuminating my bottle. Adjust the light's intensity, color, and position to achieve the desired effect.
 - The key light should create highlights and shadows that define the shape of the bottle.
 - Rim light :Place this light behind the bottle to create a rim effect. This will help separate the bottle from the background and add depth.
 - Adjust the intensity to ensure it highlights the edges of the bottle without overwhelming the scene.

2. Analyze the Project Needs Regarding Detail Quality, Materials, Lighting, and Animation

- **Lighting:**(Arnold light)
 - **Lighting Setup:** Plan the lighting setup for rendering. This may include the Area Light or Spotlight from the Arnold lights. to create a visually appealing presentation of the bottle.
 - **Fill Light :**this light opposite the key light to soften shadows created by the key light.
 - Lower the intensity of the fill light compared to the key light to ensure it fills in shadows without overpowering the key light.
- **Reflection and Refraction:** Consider how the glass material will interact with light, including reflections and refractions that enhance realism.

2. Analyze the Project Needs Regarding Detail Quality, Materials, Lighting, and Animation

- **Lighting:**(Arnold light)
- **Render Settings**
 - Make sure to set Arnold as your renderer in the **Render Setup** dialog.
 - Adjust the render settings for quality, such as sampling rates, to ensure the best output.
- **Test Renders**
 - Perform test renders to see how the lighting affects the appearance of the bottle.
 - Make adjustments as needed based on the results of mytest renders.

2. Analyze the Project Needs Regarding Detail Quality, Materials, Lighting, and Animation

- **Animation:(Keyframes)**
 - **Animation Requirements:** It will include animation of curtain , and movement of camera around perfume and movement of petals of flower (may include path animation for them)
 - **Timing and Flow:** range of frames will be between this range (300 to 900 frames).depend on movement and smoothing of animation